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SUCCESSFULL HEART SURGERY
IN A 800 GM PRETERM BABY AT GKNM HOSPITAL

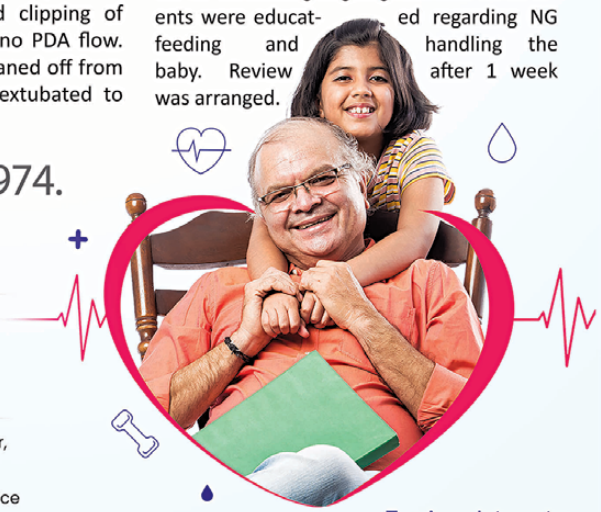
Patent ductus arteriosus (PDA) is a persistent communication between the descending thoracic aorta and the pulmonary artery that results from failure of normal physiologic closure of the fetal ductus (DA) soon after birth. While the DA is an essential component of the fetal circulatory system, patent ductus arteriosus is associated with significant neonatal morbidity and mortality. PDA is one of the most common congenital heart defects, accounting for 5%–10% of all congenital heart disease in term infants. Incidence is much higher in preterm infants (20%–60%) in comparison with term infants. Symptomatic PDA is present in 48% of infants with a birth weight of less than 1Kg. There are three treatment options for PDA: fluid restriction, pharmacologic intervention (Paracetamol or Ibuprofen) and surgical ligation. Surgical ligation is typically reserved for preterm patients who have failed multiple pharmacological treatment attempts.

I would like to highlight the challenges we faced while managing a very preterm precious female baby (conceived after 12 years of married life) and one of the surviving twins, at our Paediatric Cardio-Thoracic Intensive care unit of GKNM Hospital.

The baby was born by emergency C-Section at 29 weeks with a birth weight of 800 grams in a tertiary neonatal care unit. The baby was intubated and given surfactant immediately after birth due to respiratory distress. During the first week of life baby had Respiratory distress syndrome, newborn jaundice and suspected sepsis which got resolved. However, the baby's respiratory distress persisted and had features of congestive cardiac failure. ECHO showed a Large Patent Ductus Arteriosus (PDA) for which she had three courses of medical treatment, all of which failed.

Therefore, the baby was referred to our centre on 23 days of life for surgical closure of hemodynamically significant PDA.
She was taken up for surgery after being medically managed for suspected sepsis, apnoea of prematurity and congestive cardiac failure. Her nutrition was optimised with Total Parenteral Nutrition (TPN) via a PICC line (peripherally inserted central catheter). Parents were fully counselled about the risks involved. The baby underwent Left Thoracotomy and clipping of PDA. Post-op ECHO showed no PDA flow. Postoperatively baby was weaned off from mechanical ventilation and extubated to Nasal NIV, the next day.

We made sure optimal oxygen supplementation was provided as the baby was at high risk of retinopathy of prematurity. ROP screening done before discharge showed zone 2 immature vessels in both eyes. We optimised calorie requirements by continuing TPN feeds post-op and slowly introduced enteral feeds until fully tolerated because of the risk of NEC (necrotising enterocolitis). The baby had frequent apnoeic episodes due to prematurity, which was managed with caffeine supplementation, careful monitoring and stimulation.
She had mild elevation of renal parameters which was managed conservatively and got normalised soon. Meticulous hand hygiene was practised while handling the baby and while using PICC line. As expected parents were very anxious, and we kept them updated regularly and encouraged to get involved in baby care.Baby made a steady recovery. She was saturating normally on room air and established full enteral feeds (on nasogastric tube feed) while discharged back to the referring tertiary neonatal hospital on POD 11 of life for continuing ongoing neonatal care.Parents were educated regarding NG feeding and handling the baby. Review after 1 week was arranged.



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